Public Economics and Development

Week 10: Types of public services: Cash vs. In-kind, Conditional vs. Unconditional

Seung-hun Lee

Taipei School of Economics
National Tsing Hua University

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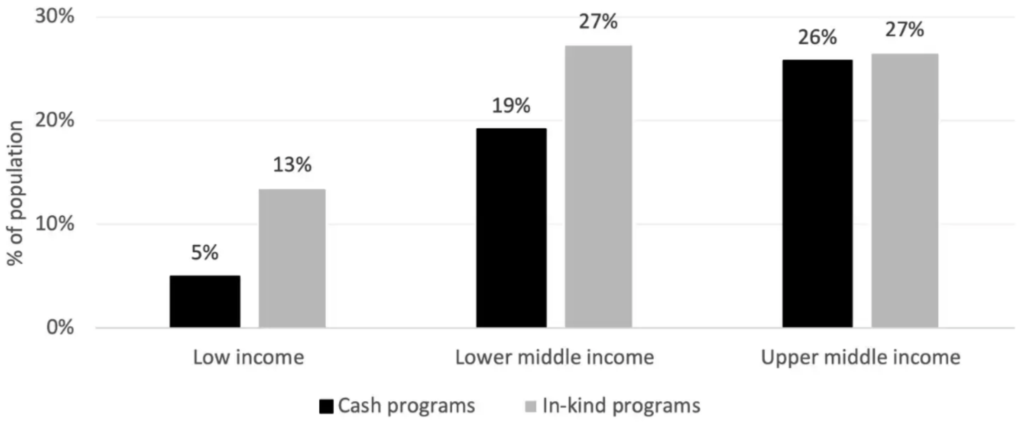
Widespread usage of Cash and In-kind transfers



Source: Gentilli, U. (2023) “Why does In-kind Assistance Persist when Evidence Favors Cash Transfers?”, Brookings Institution

Cash vs. in-kind transfers

Case for cash

- Recipients know their own needs best – cash respects consumer sovereignty
- Less administrative cost (no procurement, storage, or distribution)
- Recipients prefer the cash equivalent and reallocate accordingly than in-kind

Case for in-kind

- Paternalism: recipients may not spend cash on socially desired goods
- Self-targeting: in-kind goods are less attractive to the non-poor
- Promote consumption of goods generating positive externality

Key question

$\rightarrow$ Respect recipient preferences, or promote socially desired consumption?

$\rightarrow$ Targeting or insurance value of in-kind transfers $>$ Higher cost relative to cash?

Conditional cash transfers: Progresa in Mexico

Mexico's Progresa (1997-2019) as the pioneering large-scale CCT

- Targets poor rural households via proxy-means testing
- Transfers conditional on school attendance (85%+) and health clinic visits
- By 2004, covered 5 million households — around a quarter of Mexico's population
- Enrollment, child health , household consumption $\uparrow$ (effect wiped out post program)
- Spawned a global wave: Used in 2 countries in 1997 $\rightarrow$ 63 LMICs in 2016

Why Progresa became a blueprint

- One of the first programs to be rigorously evaluated via randomized rollout
- Demonstrated simultaneous gains in enrollment, health, and consumption
- Spawned a large literature debating on optimal design
    - How much do conditions matter?
    - Who benefits most?
    - What is the optimal CCT/UCT mix?

Unconditional cash transfers: GiveDirectly in Kenya

An NGO-driven large-scale unconditional cash transfer

- Rolled out in poor households in rural Kenya via M-Pesa mobile money
- Eligibility based on a simple means test (thatched roof as poverty proxy)
- Average transfer of $709 PPP, delivered as lump sum or monthly installments
    - Roughly two years of per capita expenditure,
- Became the benchmark against which other transfer modalities are compared

What the evidence shows

- Monthly household consumption from $158 to $193 PPP nine months after transfer
- Significant gains in assets, food security, and psychological well-being
- No evidence of wasteful spending (alcohol, tobacco)
- Then, is there a need to condition benefits at all?

The CCT vs. UCT debate

Case for CCT

- Correct underinvestment in human capital due to misperceived returns
- Imposing a condition deters the non-needy from applying
- Taxpayers accept conditioned redistribution more readily

Case for UCT

- More autonomy for households
- Conditions distort choices: households may comply not because it is optimal but because it is required
- Simpler and cheaper to administer (no need to monitor compliance)

Key breaking point

→ If the goal is welfare improvement, do conditions add beyond the income transfer?

→ If the goal is behavior change, are conditions the most efficient way to achieve it?

Roadmap for today

1. Overview setup of the topic

2. Cash vs In-kind Transfers
   2.1 Hidrobo, Hoddinott, Peterman, Margolies, Moreira (2014): Cash, Food or Vouchers? Evidence from Randomized Experiment in Northern Ecuador
   2.2 Cunha, De Giorgi, Jayachandran (2019): The Price Effects of Cash vs In-Kind Transfers
   2.3 Gardenne, Norris, Singhal, Sukhtankar (2024): In-kind Transfers as Insurance

3. Conditional vs. Unconditional transfers
   3.1 Baird, McIntosh, Özler (2011): Cash or Condition? Evidence from a Cash Transfer Experiment
   3.2 Bergstrom, Dodds (2021): The Targeting Benefit of Conditional Cash Transfer

Hidrobo, Hoddinott, Peterman, Margolies, Moreira (2014): Cash, Food or Vouchers? Evidence from Randomized Experiment in Northern Ecuador

Q: How does form of assistance matter for welfare outcomes?

Debate over the form of assistance has long history

- Proponents of cash
  - Autonomy: Allow beneficiaries to use transfers as they see fit
  - Less stigma: Less visible compared to in-kind transfers
- Proponents of in-kind transfers
  - Targeting: Only those who truly need it will take-up these benefits
  - Paternalism: In-kind transfers facilitate changes in a particular behavior

This paper uses RCT to comparatively evaulate the effects of cash and in-kind transfers

- Effect of cash, food vouchers, and food transfers on food consumption
- Track quantity and quality (calory intake, dietary diversity)
- Also provides analysis on cost-effectiveness across different programs

Program design

Background: WFP assistance to poor households and Colombia refugees in Ecuador

- 6 Monthly transfers of cash, food, and vouchers $\simeq$ \$40 (May - Oct 2011)
- Eligibility determined by proxy means test based on surveys
- Cash transfers sent to ATM cards and did not expire
- Vouchers redeemable for nutritionally-approved foods at markets (nontransferrable)
- Food transfers gave rice, oil, lentiloes, canned sardines

Some characteristics of the recipients

- Mostly women: Original intent was to increase women's role in household decisions
- Mostly poor: Dirt floors, rarely own TV/computer/automobiles
- Has 1-2 children (mostly aged between 6-15)
- All received same lecture on significance of nutrition

Research design

Experiment details

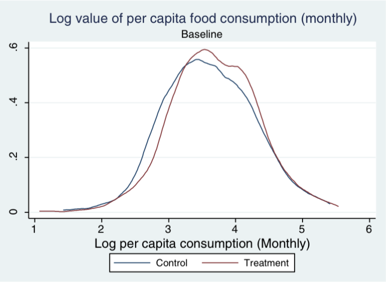
- Randomization at neighborhood level
- Neighborhoods randomized into one of the four treatments
- Thus, treatment arms are comparable at baseline
- Regression based on OLS (Analysis of Covariance)

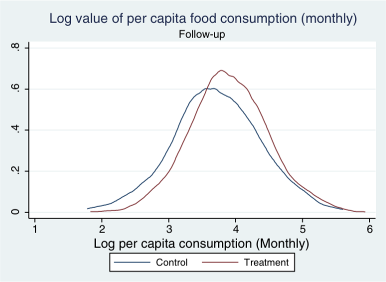
$$Y_{hj,1} = \beta_f \text{food}_j + \beta_c \text{cash}_j \beta_v \text{voucher}_j + \gamma Y_{hj,0} + \text{StrataFE} + \text{NeighborFE} + e_{hj}$$

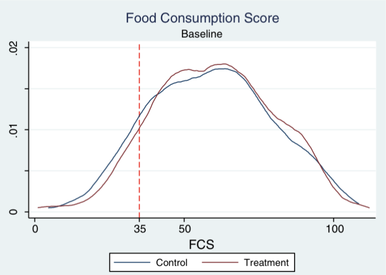
Additional data obtained

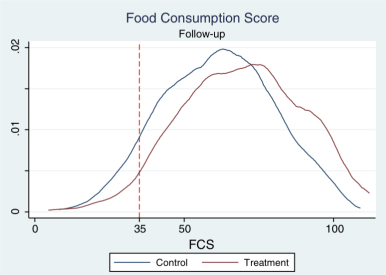
- Following quality measures are used
  - Dietary diversity (DDI): Sum of no. of distinct food items consumed last week (1-40)
  - Household dietary diversity (HDDS): no. of distinct food groups consumed (1-12)
  - Food cosumption score (FCS): $\simeq$ HDDS, different weights per food group (35 as cutoff)
- Caloric intake constructed from Nutritional Database for Standard Reference (USDA)

Overall improvements across treatment groups









All treatments show increases in quantity of consumption

	LEVELS			LOGS		
	Food consumption (per capita)	Non-food consumption (per capita)	Total consumption (per capita)	Food consumption (per capita)	Non-food consumption (per capita)	Total consumption (per capita)
Food treatment	9.22	9.22	18.50	0.20	0.15	0.17
	(2.79)***	(3.30)***	(5.02)***	(0.04)***	(0.07)**	(0.05)***
Cash treatment	5.47	6.81	12.66	0.14	0.07	0.11
	(2.56)**	(3.93)*	(5.09)**	(0.04)***	(0.06)	(0.04)***
Voucher treatment	6.38	6.78	13.45	0.15	0.13	0.13
	(2.58)**	(2.82)**	(4.38)***	(0.04)***	(0.06)**	(0.04)***
R^2	0.21	0.17	0.22	0.26	0.25	0.25
N	2087	2087	2087	2087	2087	2087
Baseline Mean	47.54	64.29	111.83	.	.	.
P-value: Food = Voucher	0.31	0.46	0.33	0.23	0.75	0.40
P-value: Cash = Voucher	0.73	0.99	0.88	0.80	0.35	0.63
P-value: Food = Cash	0.17	0.57	0.30	0.14	0.27	0.21

Food transfer $\rightarrow$ Calorie intake $\uparrow$ and Voucher $\rightarrow$ DDI

	Log caloric intake (per capita)	HDDS	DDI	FCS	Poor food consumption
Food treatment	0.21	0.61	2.36	6.96	-0.05
	(0.04)***	(0.12)***	(0.44)***	(1.22)***	(0.02)***
Cash treatment	0.12	0.47	2.64	6.57	-0.02
	(0.04)***	(0.11)***	(0.42)***	(1.29)***	(0.02)
Voucher treatment	0.18	0.60	3.13	9.56	-0.04
	(0.04)***	(0.12)***	(0.45)***	(1.39)***	(0.02)***
R^2	0.17	0.16	0.27	0.16	0.08
N	2087	2087	2087	2087	2087
Baseline Mean	1895.43	9.18	17.27	59.86	0.11
P-value: Food = Voucher	0.40	0.86	0.07	0.07	0.73
P-value: Cash = Voucher	0.15	0.16	0.22	0.05	0.13
P-value: Food = Cash	0.03	0.12	0.48	0.77	0.09

Cost-effectiveness and remainig discussion

Estimates of cost-effectiveness

- Costs: WFP's accounting ledgers
  - Modality-specific staff costs, debit card production, storage facilities etc
- Costs: Food transfers $>$ Cash $\simeq$ Vouchers
  - For each transfer: \$11.46, \$ 2.99, and \$3.27 each
  - For raising certain outcomes, Food transfers $>$ Cash $>$ Vouchers
- Recipient preference for cash: 9% wish not to receive further help vs. 30% in others

Discussion

- Choosing the best mode of assistance depends on specific objectives
- Simply improve welfare? Cash is preferable
- Promote particular behaviors? In-kind might be better
- Considering recipient preference? In-kind may not be best suited

Cunha, De Giorgi, Jayachandran (2019): The Price Effects of Cash vs In-Kind Transfers

Q: How does different anti-poverty program affect prices?

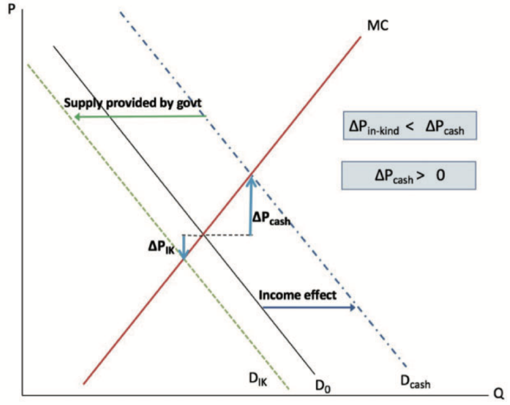
How do in-kind and cash transfers affect market circumstances?

- Cash transfers: Increase demand for normal goods
  - Think of how expansionary policy affects price levels
- In-kind: Increase demand for normal goods, but also increase supply
- Potentially have different effects depending on competition structure
  - Perfect competition: Shift wealth between buyers and sellers
  - Imperfect competition: Correct inefficiencies by reducing high prices

This paper tests the presence of these price effects in rural Mexico

- Compare between status quo, cash transfers, and in-kind transfers
- Estimates changes in prices following inflow of support
- Disaggregates size of price effects across villages with different development

Conceptual framework: Perfect comptition



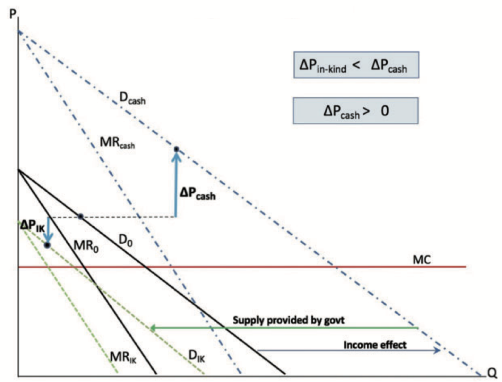
Cash transfers

- Demand curve shift right ($\because$ income $\uparrow$)
- Hence, price $\uparrow$

In-kind transfer (equivalent value)

- Demand curve shift right ($\because$ income $\uparrow$)
- Some of the demand provided by govt
- Residual demand for local suppliers shift left
- Hence, price $\downarrow$
  - In-kind good demand do not rise drastically
  - Unless it is a luxury good

Conceptual framework: Imperfect comptition



Assume N firm Cournot-Nash competition

- Constant MC at c, Demand $p = d - Q$
- Equilibrium $p = \frac{d+Nc}{N+1}$

Cash transfers

- Demand curve shift right ($\because$ income $\uparrow$)
- Hence, price $\uparrow$

In-kind transfer (equivalent value)

- Demand curve shift right ($\because$ income $\uparrow$)
- Residual demand for local suppliers shift left
- Hence, price $\downarrow$
- Room to correct excessively high price

Experiment design and context

Programa de Apoyo Alimentario (PAL)

- 5,000 small ($< 2,500$ ppl) highly marginalized villages
- Monthly allotment of basic food items (rice, beans etc) and supplements (tuna etc)

Experiment design

- Randomly selected 208 villages
- Assigned to In-kind, Cash transfer, and control group
- Value of cash is 150 MXP: 18% of food expense & 8% of income per month
- Value of in-kind is roughly 206 MXP, but w/ assumptions, have similar value to cash
    - 116 MXP is consumed
    - 35 MXP of extra consumption of transferred goods, 55 MXP not consumed
    - The latter two discounted to 2/3 $\rightarrow$ in-kind value of 146 MXP
- Uses household and store surveys to obtain data on price

Price in in-kind fell relative to cash villages

	All PAL goods	Basic PAL goods only	All PAL goods	Basic PAL goods only	All PAL goods	Basic PAL goods only
Outcome =	price	price	price	price	Δprice	Δprice
	(1)	(2)	(3)	(4)	(5)	(6)
In-kind	−0.037*	−0.033	−0.036*	−0.033	−0.062**	−0.025
	(0.020)	(0.020)	(0.020)	(0.020)	(0.029)	(0.024)
Cash	0.002	0.014	0.003	0.012	0.000	0.039
	(0.023)	(0.027)	(0.023)	(0.026)	(0.031)	(0.029)
Lagged normalized unit value	0.027	0.127***				
	(0.021)	(0.042)				
Observations	2,335	1,617	2,335	1,617	2,335	1,617
***Effect size*: In-kind - Cash**	**−0.039****	**−0.047****	**−0.038****	**−0.045****	**−0.063****	**−0.064****
***H_0*: In-kind = Cash (*p*-value)**	**0.02**	**0.04**	**0.03**	**0.04**	**0.02**	**0.02**

Price differences larger in less developed villages

	Below-median development	Above-median development	All villages	Villages with market power	Villages without market power	Below-median price correlation	Above-median price correlation
Outcome :	price	price	price	price	price	price	price
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
In-kind	-0.036	-0.033	-0.033	-0.048*	-0.005	-0.060**	-0.019
	(0.031)	(0.027)	(0.021)	(0.025)	(0.021)	(0.028)	(0.028)
Cash	0.015	-0.007	-0.007	0.007	-0.005	0.002	-0.014
	(0.032)	(0.037)	(0.032)	(0.029)	(0.026)	(0.032)	(0.032)
Development index below-median x In-kind			-0.006				
			(0.018)				
Development index below-median x Cash			0.018				
			(0.031)				
Market power village x In-kind							
Market power village x Cash							
Price correlation below-median x In-kind							
Price correlation below-median x Cash							
Observations	1,094	1,210	2,304	1,733	602	1,115	1,220
***Effect size:* In-kind - Cash**	**-0.051****	**-0.027**	**-0.026**	**-0.055*****	**0.000**	**-0.063*****	**-0.006**
***H_0:* In-kind = Cash (p-value)**	**0.02**	**0.37**	**0.37**	**0.01**	**1.00**	**0.01**	**0.81**

Other findings and key takeaways

Other findings

- The price effects are actually persistent
  - Villages treated early had price effects even when later-treated villages began treatement
- Purchasing power in developed villages do not change much
- Larger price effect in poor villages due to less market integration and competition
- Local supplier profit increases in cash transfer villages (no change in in-kind village)

Discussions

- In-kind transfers can correct prices set excessively high due to imperfect competition
- But if government is inefficient, this effect could be limited
- If so, cash transfers with policies to promte supply-side competition may be preferred
- Choice of cash vs. in-kind depends on policy goals and market circumstances

Gardenne, Norris, Singhal, Sukhtankar (2024): In-kind Transfers as Insurance

Q: How does in-kind transfers insurce against price volatility?

Low-income households are vulnerable to price variations

- Chages in price of key consumption goods affect stability of their budget
- Particularly for developing countries: Market integration $\downarrow$ and weather shocks
- If households only receive cash transfers, they cannot insure against such shock
- The insurace aspect of transfers, not jus transfers, is understudied

This paper sheds light on the insurace aspect of in-kind transfers

- Conceptual framework of price-indexed transfers equalizing marginal utility of income
- In-kind transfers can provide additional insurance value on top of redistrubution
  - Especially when price-indexing is difficult due to lack of high-frequency data
- Empirically tests the idea using calorie intake as proxy for marginal utility of income
  - Using a flagship in-kind transfer program in India (Public Distribution System, PDS)

Conceptual Framework

Optimal insurance policy

- Consider households consuming several goods, where one of its price p_j varies
- It maximizes indirect utility $v(p, y)$ (y: income)
- Insurance: Price-dependent transfers $x(p_j)$
- Optimal insurance equates marginal utility of income (μ) in all states of the world

$$v_y(p, y + x(p_j)) = \mu$$

- If v_y (MU of income) increase in price, then $x(p_j)$ increases in price

Role of in-kind transfers (vs. benchmark cash transfer)

- Same goods in fixed quantity vs same expected monetary value across all states
- $p_j \uparrow \rightarrow$ Market value of an in-kind transfer $\uparrow \rightarrow$ purchasing power $\uparrow$
- In-kind transfers have insurance value (price-indexed transfers) – not fixed transfers
- This hinges on whether additional income more valuable when prices are high

Empirical testing

India and PDS system

- Prices are volatile and many fail to meet caloric requirements
- PDS provides rice and other staple foods at below-market prices

Transferring theory to data

- MU of income proxied with indicator for failing to meet calorie requirements
  - Assume: Likelihood of failing to meet requirments $\uparrow$ associated with MU of income $\uparrow$
  - If unlikely to meet requirement, additional income valuable for meeting this requirement
- Relevant price and calorie intake data from National Sample Survey (NSS, 2003-12)
- Also control for household characteristics (obtained from NSS)
- Uses the following regression to estimate relationship between price and caloric intake

$$c_{it} = \beta \text{Riceprice}_{-i,t} + X_{it}\lambda + \delta_{da} + \theta_t + \phi_{\text{round}} + e_{it}$$

where leave-out price is used to net out influence of own-taste

More likely to not meet requirements with higher prices

	All districts				RPS districts	
	(1)	(2)	(3)	(4)	(5)	(6)
log market rice price	−0.131	−0.112	−0.131	−0.165	−0.301	−0.304
	(0.041)	(0.044)	(0.041)	(0.042)	(0.078)	(0.081)
District-sector fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
District-sector-season fixed effects	Yes	Yes	No	Yes	Yes	Yes
Period fixed effects	Yes	Yes	Yes	No	Yes	Yes
HH controls	Yes	No	Yes	Yes	Yes	Yes
SES controls	Yes	No	Yes	Yes	Yes	Yes
Observations	524,911	524,911	524,911	524,911	175,065	175,065

Particularly for poor rural households

	By median SES		By census region		Rural by landowning	
	Below	Above	Rural	Urban	Landless	Landowning
	(1)	(2)	(3)	(4)	(5)	(6)
log market rice price	−0.234	−0.035	−0.191	−0.024	−0.302	−0.157
	(0.056)	(0.039)	(0.053)	(0.058)	(0.090)	(0.049)
Equality of effect (*p*-value)		0.00		0.03		0.09
Observations	211,796	313,115	316,234	208,677	63,614	252,620

Also reduces sensitivity of meeting requirements to market prices

	Meets MCR			log calories per capita		
	All	Below-median SES	Above-median SES	All	Below-median SES	Above-median SES
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. IV of outcomes on PDS value						
PDS value (100 ₹)	0.107	0.136	0.078	0.064	0.062	0.069
	(0.052)	(0.063)	(0.048)	(0.040)	(0.039)	(0.041)
Equality of effects (p-value)			0.049			0.690
Effective F-stat	19.07	17.03	16.37	19.07	17.03	16.37
10 percent bias crit. val.	18.66	17.84	18.87	18.65	18.15	18.85
Panel B. IV of outcomes on PDS value						
log market rice price	−0.260	−0.468	−0.123	−0.166	−0.260	−0.106
	(0.054)	(0.086)	(0.044)	(0.033)	(0.057)	(0.031)
Market rice price × PDS value	0.177	0.209	0.274	0.150	0.143	0.247
	(0.066)	(0.075)	(0.105)	(0.049)	(0.045)	(0.073)
Equality of effects (p-value)						
log market rice price			0.000			0.011
Market rice price × PDS value			0.438			0.027
Pred. rice elasticity at mean PDS	−0.207	−0.384	−0.071	−0.122	−0.203	−0.059
	(0.051)	(0.094)	(0.036)	(0.033)	(0.058)	(0.027)
Mean PDS value	0.30	0.40	0.19	0.30	0.40	0.19
SD PDS value	0.604	0.668	0.512	0.604	0.668	0.512
First percentile PDS value	0.000	0.000	0.000	0.000	0.000	0.000
Ninety-ninth percentile PDS value	2.556	2.685	2.325	2.556	2.685	2.325
Observations	524,911	211,796	313,115	524,911	211,796	313,115

Key takeaways and discussion

Further findings

- Value of in-kind that would ensure meeting caloric requirements
  - Find x s.t. $-0.260 + 0.177 \times x = 0$
  - This x is 147 Rupees, 1.5 times the average nonzero transfer
- Thus, PDS implemented in the sample period did contribute to substantial insurance

Policy implications

- So does in-kind transfers dominate cash transfers? Not necessarily
- Need to consider implementation costs (As discussed in the previous paper)
- Nevertheless, insurance value provided by in-kind could be worthwhile for recipients
- Especially when recipients are exposed to price risk that affects their welfare

Baird, McIntosh, Özler (2011): Cash or Condition? Evidence from a Cash Transfer Experiment

Q: How to conditioning benefits affect recipient behaviors?

Relative merits between conditioning (CCT) and not conditioning (UCT) cash transfers?

- Evidence that both CCT and UCT do affect behaviors across different settings
- Proponents of CCT
  - Correct for underinvestment in education and health by imposing conditions
  - Make transfers politically acceptable to the non-poor
- Proponents of UCT
  - Marginal contribution of conditioning unknown
  - Conditioning may be difficult to implement in dedveloping countries

This paper examines marginal contribution of conditioning cash transfers

- Cash transfers to households with school-age girls in Malawi
- RCT where some receive CCT, others are under UCT, and remaining receive none
- Examine behaviors that CCT conditions (education)
- But also investigate others (human capital, underage marriage / pregnancy)

Experimental setting and design

Zomba District in Malawi

- One of the poorest in the world (2008 GNI per capita is $760)
- Enrollment to secondary school is low (24%)
- 176 Enumeration Areas spread across urban, near rural, far rural areas
- Surveyed to identify never-married females aged 13-22

Treatment assignment

- CCT arm (T1, 46 EAs): Monthly transfer conditional on regular school attendance
  - Transfer amount determined by random: $4 - $10 per month
  - Separate payment to schoolgirls ($1-$5) to cover school fees
- UCT arm (T2, 27 EAs): No school requirement, same financial offer
- Remaining EAs (88) are in control group
- Track attendence, performance, marriage, pregnancy for '08 and '09
- Potential concern: Selective attrition (especially for control group)

No selective attrition!

TABLE I
ANALYSIS OF ATTRITION

	Dependent variable					
	(1)	(2)	(3)	(4)	(5)	(6)
	=1 if surveyed in Round 3	=1 if surveyed in all three Rounds	=1 if took educational tests	=1 if information found in Round 2 survey	=1 if information found in Round 3 school survey	=1 if legible ledger found
Conditional treatment	0.020	0.021	0.029*	0.033	−0.000	0.116*
	(0.015)	(0.030)	(0.016)	(0.024)	(0.027)	(0.064)
Unconditional treatment	0.021	0.030	0.035*	−0.029	0.014	0.061
	(0.019)	(0.024)	(0.020)	(0.053)	(0.028)	(0.077)
Mean in the control group	0.946	0.893	0.929	0.890	0.935	0.378
Number of observations	2,284	2,284	2,284	2,284	983	821
Prob > F(Conditional = Unconditional)	0.965	0.797	0.801	0.246	0.627	0.513

Higher enrollment in CCT (based on teacher reports)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Year 1: 2008			Year 2: 2009			Year 3: 2010	
	Term 1	Term 2	Term 3	Term 1	Term 2	Term 3	Total terms (6 terms)	Term 1, post-program
Panel A: Program impacts on *self-reported* school enrollment								
Dependent variable: =1 if enrolled in school during the relevant term								
Conditional treatment	0.007	0.019*	0.041**	0.049***	0.056***	0.061***	0.233***	0.005
	(0.011)	(0.011)	(0.017)	(0.017)	(0.018)	(0.019)	(0.070)	(0.025)
Unconditional treatment	0.034***	0.051***	0.054***	0.072***	0.095***	0.101***	0.406***	0.074***
	(0.010)	(0.011)	(0.018)	(0.021)	(0.022)	(0.021)	(0.079)	(0.026)
Mean in the control group	0.958	0.934	0.900	0.831	0.800	0.769	5.191	0.641
Number of observations	2,087	2,087	2,086	2,087	2,087	2,087	2,086	2,086
Prob > F(Conditional = Unconditional)	0.006	0.012	0.460	0.299	0.102	0.098	0.038	0.028
Panel B: Program impacts on *teacher-reported* school enrollment								
Conditional treatment	0.043***	0.044***	0.061***	0.094**	0.132***	0.113***	0.535***	0.058*
	(0.015)	(0.016)	(0.018)	(0.041)	(0.035)	(0.039)	(0.129)	(0.033)
Unconditional treatment	0.020	0.038**	0.018	0.027	0.059	0.033	0.231*	0.001
	(0.015)	(0.017)	(0.023)	(0.038)	(0.037)	(0.039)	(0.136)	(0.036)
Mean in the control group	0.906	0.881	0.852	0.764	0.733	0.704	4.793	0.596
Number of observations	2,023	2,023	2,023	852	852	852	852	847
Prob > F(Conditional = Unconditional)	0.173	0.732	0.067	0.076	0.014	0.020	0.011	0.108

Better educational performance from CCT

	Dependent variable			
	(1)	(2)	(3)	(4)
	English test score (standardized)	TIMMS math score (standardized)	Non-TIMMS math score (standardized)	Cognitive test score (standardized)
Conditional treatment	0.140***	0.120*	0.086	0.174***
	(0.054)	(0.067)	(0.057)	(0.048)
Unconditional treatment	−0.030	0.006	0.063	0.136
	(0.084)	(0.098)	(0.087)	(0.119)
Number of observations	2,057	2,057	2,057	2,057
Prob > F(Conditional= Unconditional)	0.069	0.276	0.797	0.756

Less early marriage and pregnancy in UCT

	Dependent variable			
	(1)	(2)	(3)	(4)
	=1 if ever married		=1 if ever pregnant	
	Round 2	Round 3	Round 2	Round 3
Conditional treatment	0.007	−0.012	0.013	0.029
	(0.012)	(0.024)	(0.014)	(0.027)
Unconditional treatment	−0.026**	−0.079***	−0.009	−0.067***
	(0.012)	(0.022)	(0.017)	(0.024)
Mean in the control group	0.043	0.180	0.089	0.247
Number of observations	2,087	2,084	2,086	2,087
Prob > F(Conditional = Unconditional)	0.024	0.025	0.265	0.003

Key findings and discussion

Other key findings

- UCT impacts on marriage and pregnancy entirely driven by girls who dropped out
  - Effects of UCT on these outcomes null among those enrolled
- Authors do a follow-up (paper in 2019 at *Journal of Development Economics*)
  - Besides sustained enrollment, effects of CCT vanish
  - Health gains of UCT besides height-for-age vanish
- Effects of CCT similar across different amounts of transfers

Discussion

- CCT can be cost-effective in raising educational attainment
- But UCTs still can improve other outcomes (health, delayed marriage)
- CCT could improve outcomes that hinge on complying to conditions (e.g. test scores)
- UCT effective if many noncompliers change behaviors following income support

Bergstrom, Dodds (2021): The Targeting Benefit of Conditional Cash Transfer

Q: Does conditional transfers have a targeting *benefit*?

Major opposition to conditional cash transfer (CCT): Targeting cost

- CCTs are conditional on investments in children's human capital (schooling or health)
- Low-income households find these conditions costly $\rightarrow$ excluded (targeting cost)
- Thus, many prefer UCTs (no targeting costs) to alleviate poverty

This paper argues that CCTs have targeting benefits that could offset targeting costs

- Sending a chilren: lumpy investment where child income is the opportunity cost
- CCTs direct money to households forgoing child income (targeting benefit)
- Provides a theoretical framework capturing targeting benefits and costs
- Proposes a optimal trade-off between CCT and UCT with estimates from rural Mexico

Theoretical framework

Set-up of the household optimziation

- Parents decide whether to send children to work (for income y_c) or to school
  - Schooling HH forgo y_c but receive higher income when children become adults
- All receive UCT t_u, schooling HH receive t_c as well (along with parental income y)
- Cannot borrow against child's future income (Otherwise, always consumption more)
- Thus, they solve (s: schooling, μ: child's ability)

$$\max_{s\in\{0,1\}} u(c) + v(\mu, s) \ \text{(s.t.)}\ c = y + y_c(1-s) + t_u + t_c s$$

Social planner problem

- Utilitarian: Maximizes sum of social welfares of schooling & non-schooling HH
- Distribute fixed budget to predetermined set of households in UCT or CCT
- Can observe s (not y, or μ) - so that CCTs goes to schooling households ($s = 1$)

Key takeaways from the framework

Conclusions: $(1 + \frac{\partial t_u}{\partial t_c})\overline{u_c(c(1))}S + \frac{\partial t_u}{\partial t_c}\overline{u_c(c(0))}(1 - S) = 0$

- Targeting benefit: Net welfare to schooling HH when t_c increases
- Targeting cost: Net welfare loss from decreasing t_u due to rise in t_c
- Key component: MU of consumption higher in schooling households!
  - Especially if y_c is large and parental income is largely equal
  - May not hold if schooling households are rich to begin with (most developed countries)
- In such case, better to increase budget to CCT

How to capture from empirically observable objects (sufficient statistics)?

- S: Share of schooling households (obtainable from data)
- $\overline{u_c(c(s))}$: Consumption distribution across households + curvature of utility functions
- $\frac{\partial t_u}{\partial t_c}$: Budget trade-off (Not observable on its own)
  - Need income effect of UCT (I), price effect of CCT (P), and S
  - I tracks how S changes with UCT, P tracks how S changes with CCT

Taking the theory to the data

Progresa program in Mexico

- Largest component: nutritional subsidy and education grant
- 320 treated localities (1998 –), 186 control localities (2000 –)
- Transfers conditional on children attending school 85%+ of days (grades 3–9)

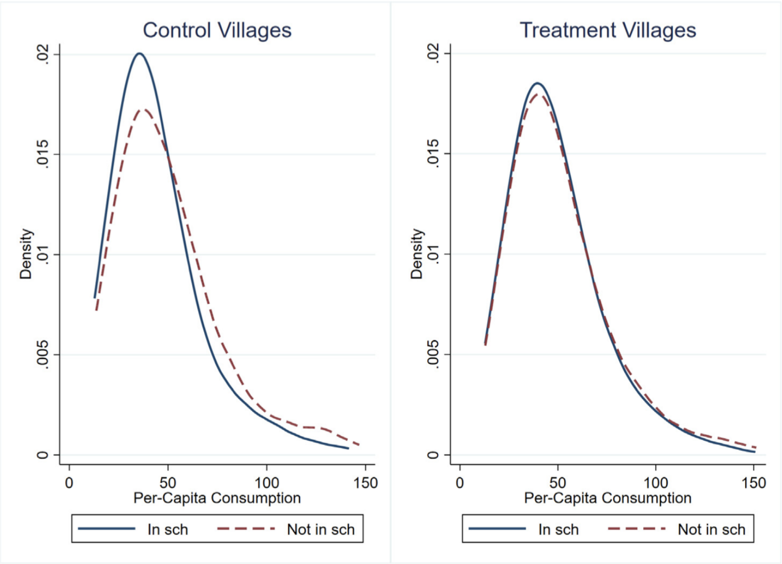
Identifying income effects (I)

- Use variation in transfers to siblings under age 12
- Enrollment below age 12 is near 100% (practically a UCT)
- Variation in sibling composition provides exogenous variation in size of t_u

Identifying price effects (P)

- Introduction of CCTs was randomized at the locality level
- Transfer schedule varies by grade and gender, providing variation in size of t_c

CCTs can narrow the consumption gap



$P = 0.008, I = 0.004$

	(1) OLS	(2) IV	(3) IV	(4) IV
t_c	0.00838***	0.00796***	0.00807***	0.00779***
	(0.002)	(0.002)	(0.002)	(0.002)
t_u	0.00372*	0.00377*	0.00385+	0.00383+
	(0.002)	(0.002)	(0.002)	(0.002)
y^f	−0.000236	0.00345	0.00285	0.00286
	(0.000)	(0.004)	(0.004)	(0.004)
$t_c^{(12+)}$				0.000920
				(0.002)
Time-varying sibling controls	N	N	Y	Y
$\mathbb{E}[\text{Enroll}\mid\text{control}, \text{year} > 97]$	0.683	0.683	0.683	0.683
$\mathbb{E}[\text{Enroll}\mid\text{treat}, \text{year} > 97]$	0.767	0.767	0.767	0.767
$\mathbb{E}[t_c\mid\text{treat}, \text{year} > 97]$	6.897	6.897	6.897	6.897
$\mathbb{E}[t_u\mid\text{treat}, \text{year} > 97]$	2.867	2.867	2.867	2.867
$\mathbb{E}[\text{famsize}]$	6.609	6.609	6.609	6.609
Observations	21321	21321	21321	21321
R-squared	0.153	0.131	0.140	0.140

- 1 Peso increase in CCT $\rightarrow$ raise schooling by 0.8 p.p
- 1 Peso increase in UCT $\rightarrow$ raise schooling by 0.4 p.p

Optimal CCT/UCT mix: quantitative results

How much of the budget should go to a CCT on targeting grounds alone?

- Large forgone y^c and limited parental income inequality $\rightarrow$ TB is substantial
- From Progresa: Targeting benefit is 79% of the targeting cost
- TB/TC $< 1 \Rightarrow$ current CCT share is slightly too high; optimal to shift some to UCT
- CCT/UCT mix that equalizes this: 33% should go to CCT

Takeaway for policy design

- The targeting benefit is a quantitatively important advantage of CCTs
- It should not be ignored when designing cash transfer programs
- CCT vs. UCT is not a binary choice: an optimal program allocates some budget to each
- CCT preferred when $c(1) < c(0)$ and enrollment is below socially optimal
- UCT preferred when $c(1) > c(0)$ and enrollment is near optimal

Unanswered questions

Program design aspects

- Optimal program size: Needs vs fiscal sustainability and general equilbrium effects?
- Amount of support – Universal amount or vary with circumstances?

Behavioral and long-run effects

- Do transfers build long-run resilience or create dependency?
- What about life-cycle effects on labor supply, savings, and investment?
- Many papers try to look at graduation programs to study this

Room for further research

- Role of digital money: How can it close the gaps?
  - Individuals can send money back to families as remittance
  - Potential insurance role: Value of transfers adjust to real-time valuation